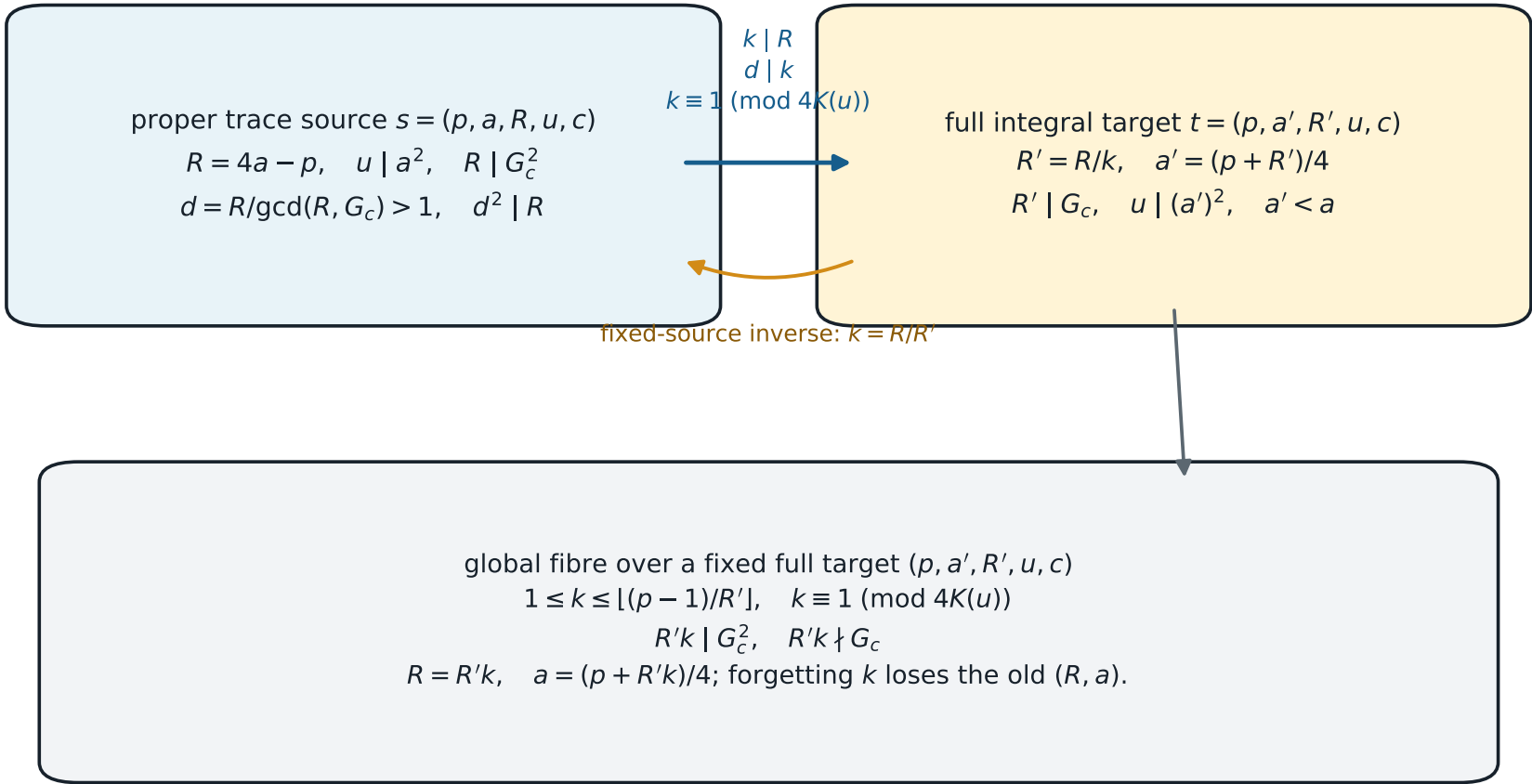


Exact same-word residual-divisor descent

Every map, congruence, inverse parameter and lost coordinate is retained.



Exact mixed-factor fixture: $p = 1,108,801$

$R = 8379 = 19 \cdot 3^2 \cdot 7^2, d = 3, u = 5, m = 20$

All positive divisors k of R containing d :

$3 \equiv 3$	$9 \equiv 9$	$21 \equiv 1$
$57 \equiv 17$	$63 \equiv 3$	$147 \equiv 7$
$171 \equiv 11$	$399 \equiv 19$	$441 \equiv 1$
$1197 \equiv 17$	$2793 \equiv 13$	$8379 \equiv 19$

Only $k = 21, 441$ satisfy $k \equiv 1 \pmod{20}$.

$k = 21 : R' = 399, a' = 277300$
 $(h', r', s', \lambda') = (5, 1, 55460, 139)$

$k = 441 : R' = 19, a' = 277205$
 $(h', r', s', \lambda') = (5, 1, 55441, 2918)$

The divisor 7 is not required by $d=3$, but its residue is required to preserve the original word $u=5$.